



Siying Wang

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Education

Phd|Renmin University of China **Information Resource Management** **2025.09-Now**

- **Research Interests:** Multi-agent Simulation, Computational Social Science, Computational Linguistics

Bachelor|Communication University of China **Big Data Management and Application** **2021.09-2025.06**

- **Major Courses:** Mathematics, Data Structures and Algorithms, Machine Learning, NLP, Knowledge Organization

Researches

First Author|《AI-medical technology integration prediction based on fine-grained temporal patent》 **2026.05**

- Use AI-medical patents of the past five years from Dawei database, identify AI and medical entities using Minimax-M3. Compute entity embeddings with bge-m3 then separately construct AI and medicine graphs. Form synonym groups from connected clusters and conduct disambiguation by synonymous node merging via LLM.
- Design half-yearly graph snapshots, calculate static graph embeddings through random walk; fuse information of two snapshots with GRU gating mechanism to generate dynamic node embeddings. Predict new links between AI and medicine entity using MLP, ablations show that snapshots and GRU promote prediction performance.

First Author|《AI-generated academic content detection based on text complexity》 **2026.04**

- For AI-generated text, use gpt2 and bert-base-uncased to construct 13 linguistic indicators from structural, semantic and linguistic complexity perspectives, based on datasets including AI-GA, CHEAT, M4 and MAGE.
- Propose an attention-based feature fusion method, use projected complexity indicators as queries to match token embeddings and obtain adjusted embedding vectors for MLP prediction, achieving an accuracy of 0.943.
- Find that human-written texts show higher complexity and weighted feature fusion is superior to linear fusion.

Co-First Author |《ARSSB: AI Research in Social Science Benchmark》 **2026.04**

- Build an end-to-end social science research pipeline with a full workflow of literature review, problem formulation, data preparation, empirical analysis, figure generation, manuscript drafting, and LaTeX typesetting. Leverage SOTA agent frameworks and LLMs to generate complete research papers on specified topics.
- Develop QA pairs to assess LLMs' foundational social science knowledge; evaluate agents' manuscripts across 5 dimensions through CLI-triggered peer review generation, structured field extraction, and discipline-specific scoring. Identify that LLMs underperform human in innovation, causal identification, and in-depth argumentation.

Internships

ByteDance **AI Data Technology Intern** **2026.01-2026.04**

- Build knowledge bases and evaluate Doubao's reasoning ability with advanced models, establish automated data table processing pipelines. Analyze cases through model-assisted tagging and manual text clustering, identify error patterns from original model responses, summarize AI capability flaws and deliver optimization proposals.
- Develop multi-turn benchmarks to address information asymmetry in clinical consultations, optimize Doctor-Environment dual-agent evaluation code and inference logic. Test mainstream models, analyze performance under varied query and guess attempts, and adopt multi-dimensional metrics including accuracy and hit rate.

Lenovo Research Institute **LLM Data Intern** **2025.03-2025.06**

- Adopting multi-processing and streaming read, reconstruct over 40GB operation trajectory data for multimodal agent desktop tasks. Leveraged GPT-4o for data augmentation, Implement image bounding box annotation, IoU based deduplication, and similarity computation, build automated Linux server-side data processing pipeline.
- Modify automated trajectory data collection scripts for web pages and APPs. Realize automated VM power on/off and snapshot rollback via Python and VMware Workstation, deploy collection environments through docker, enable data synchronization with the host machine via directory mounting, building over 1 million data samples.